

DATA QUALITY REPORT

Collections Analytics Dataset — Major Issues, Detection, Treatment, and Business Impact

The dataset arrived as seventeen relational tables spanning roughly eight months of collections activity. Its documentation warned that duplicates, conflicting timestamps, inconsistent identifiers, and schema drift had been introduced on purpose. That warning proved accurate. What follows is an account of every material defect found, how it was detected, what was done about it, and what it cost the business in distorted numbers. The central finding can be stated in one line: reported recovery of ₹134.1 crore overstates the true figure of ₹107.5 crore by 24.8 percent, and the reported month-on-month improvement dissolves once the underlying records are corrected.

A single discipline governed every decision. Rows were removed only when they were proven noise. Nothing that resembled a real account, a failed payment, an orphaned identifier, or a written-off loan was deleted, because thinning the population is precisely how a conversion rate can be made to rise without any real gain. Defects that could not be resolved by deletion were flagged and carried forward.

Scale of the cleaning

Across all tables, 639,185 raw rows reduced to 605,297 golden rows. The 33,888 rejected rows were, without exception, exact duplicates or payment retries. Payments fell from 25,500 to 23,083 after removing 486 exact duplicates and 1,931 retry references. Agents collapsed from 30,000 rows to 1,000 real agents. Borrowers resolved from 30,600 to 11,015. Calls dropped 1,271 exact duplicates to 90,079. WhatsApp events shed 600 duplicates to 60,000. Eleven further tables carried no material defect and were left intact.

Reported/Naive	Corrected/Golden
Recovery: ₹134.1 Cr	Recovery: ₹107.5 Cr
11% MoM improvement	Flat to Declining
PTP-kept: 4,489	Verified: 567
Agents: 30,000	Real agents: 1,000
Peak call hour: 23:00	Peak call hour: 05:00

Table	Raw	Rejected	Golden	Principal defect
Payments	25,500	2417	23083	486 exact duplicates + 1,931 retry references
Agents	30,000	29000	1,000	Identity collapse to 1,000 real agents
Borrowers	30600	600	11,015	600 exact dups + reused IDs
calls	91350	1271	90,079	1,271 exact dups; 3 timezones
Whatsapp events	60600	600	60,000	600 exact dups
call dispositions	35000	0	35,000	3,904 split codes (PTP vs PROMISE_TO_PAY)
account status history	60000	0	60,000	7,758 terminal-state reopenings

Issue 1 – Duplicate payments inflating recovery

Detection. Two tests ran against the payments table. The first counted rows identical across every column; 486 surfaced. The second grouped successful payments by transaction reference and counted references appearing more than once, which exposed 2,530 repeats. A retry writes a fresh row while carrying the original reference and amount forward, so a reference that appears twice with the same value is one payment recorded twice, not two payments.

Treatment. Exact duplicates were dropped. Among successful payments, one row was kept per reference, ordered by event time with the payment identifier as a deterministic tiebreaker so that re-runs select the same row every time. Reversed payments were then subtracted rather than ignored. Rows carrying a null reference could not be de-duplicated safely and were retained under a flag.

Business impact. The correction removed ₹26.7 crore of phantom recovery. Gross successful payments fell from ₹134.1 crore to ₹117.0 crore after de-duplication, and to ₹107.5 crore once ₹9.5 crore of reversals were netted out. The inflation was not constant month to month — it ranged from

roughly 22 to 28 percent — which means duplicate density alone can manufacture the appearance of a trend where none exists.

Issue 2 – Agent and borrower identity collapse

Detection. Row counts diverged sharply from distinct-key counts. The agents table held 30,000 rows but 1,000 distinct agent identifiers, and every identifier carried conflicting names, teams, and vendor codes across its rows. Borrowers showed the same pattern in milder form: 30,600 rows resolved to 11,015 identifiers, and a single borrower identifier appeared against entirely different people — one identifier mapped to both a Kolkata resident and a Hyderabad resident with different phone numbers.

Treatment. These are not duplicates and were not deleted as such. Each identifier was resolved to one golden record by keeping the most recently updated row, with tenure derived from the earliest join date rather than the latest. Agent records carry a confidence marker set to low, so that any downstream statement about individual agent performance inherits the caveat automatically.

Business impact. Any analysis of agent productivity, tenure effects, or vendor comparison rests on identifiers that do not map cleanly to people. Findings at the agent level are therefore reportable only as hypotheses. The resolution preserves every real agent and borrower while making the fragility explicit rather than hiding it inside a confident-looking average.

Issue 3 – Mixed timezones misplacing events

Detection. Accounts, calls, sessions, and vendors each carried a timezone column populated with three values in roughly equal thirds: UTC, Asia/Kolkata, and Asia/Dubai. Event timestamps were stored as local wall-clock readings. A test compared the busiest calling hour computed from raw timestamps against the same figure after conversion.

Treatment. Every event was normalised to Asia/Kolkata. Tables without their own timezone column inherited it from the parent account. Durations, which subtract two stamps in the same zone, were left untouched because the arithmetic is already correct; only absolute hour-of-day and cross-table joins required the shift.

Business impact. The apparent peak calling hour moved from 23:00 to 05:00 once the records were aligned. A staffing or best-time-to-call decision built on the raw figure would have concentrated effort five and a half hours away from the real peak. Monthly boundaries also shifted, which matters directly for any month-on-month comparison of recovery.

Issue 4 – Schema drift and split disposition codes

Detection. The dispositions table carried the code PTP and the code PROMISE_TO_PAY in parallel, both present across the legacy, v1, and v2 schema versions. A cross-tabulation of code against version confirmed the two labels describe one outcome. Vendor records and account records showed

the same version fragmentation.

Treatment. A mapping unified PTP into PROMISE_TO_PAY before any rate was computed, producing a clean disposition field alongside the original. The raw codes were retained for audit.

Business impact. 3,904 promise-to-pay dispositions had been split across two labels. Counting either label alone understates the promise-to-pay rate by close to half. The reported figure, drawn from a single label, was therefore mechanically low before any behavioural interpretation could even begin.

Issue 5 – Overwritten status history and denominator risk

Detection. The status-history table recorded up to nine status changes per account, and 7,758 accounts moved from a terminal state — paid, closed, or written off — back into an active or delinquent one. A written-off loan does not reopen. Half the table also carried a recording timestamp earlier than the event timestamp, an artifact of the same timezone mismatch rather than a true late arrival.

Treatment. The event timestamp was treated as the ordering truth and the recording timestamp discarded for sequencing. A point-in-time status was reconstructed by taking each account's latest event as of any chosen date, and the terminal reopenings were flagged. The current snapshot in the accounts table, which contradicts both the history and the outstanding balance for 7,382 supposedly paid accounts, was demoted to a reference field.

Business impact. Without point-in-time reconstruction, a conversion rate computed on the current population would silently exclude written-off accounts as they left the book, lifting the rate through attrition alone. Rebuilding the collectible population month by month held it flat near 57 percent, which rules out denominator attrition as an explanation for the reported improvement.

Further defects, resolved without deletion

Several tables carried smaller faults handled by tagging rather than removal. Accounts held 2,913 broken borrower links — 455 null and 2,458 pointing to non-existent borrowers — each marked valid, missing, or orphan and kept in place. Agent sessions contained 222 overlapping login windows that would double-count agent-hours if summed naively; these were flagged for interval merging. Daily targeting held 160 duplicate account-day rows and revealed 6,656 accounts never targeted at all, which became a natural control group for the counterfactual analysis. Calls carried 1,827 records with no agent, consistent with automated dialling, tagged rather than attributed to a person. Promise records labelled 4,489 promises as kept; only 567 had a matching payment within seven days, so an independent verification flag now sits beside the original label.

Net effect on the headline numbers

Reported recovery of ₹134.1 crore corrects to ₹107.5 crore. The claimed 11 percent month-on-month improvement corrects to a flat-to-declining trend, down roughly 5 percent from January to July. Promises labelled kept fall from 4,489 to 567 verified. Thirty thousand agent rows resolve to 1,000 real agents. The peak calling hour moves from 23:00 to 05:00.

Every figure in this report is reproducible from the raw files through the documented pipeline. The raw tables were never altered; the golden layer is a separate, versioned copy, which keeps the original defects available as evidence for the corrections claimed here.